# Cheng Dai

[Email] | [Github] | [Homepage] | [Google Scholar] | [Beijing, China]

## Overview

*I'm a researcher specializing in thermal sensing, video generation, and physical AI.*

*My research investigates how generative models can learn the dynamics, geometry, and physical structure of real-world environments, and how these models can support perception, planning, and interaction.*

## Positions

**Westlake University, School of Science** Sep. 2025 - Jul. 2026

*Research Assistant* Supervisors: Xin Yuan and Fanglin Bao

· Dissertation Fields: HADAR (nature2023, Temperature-emissivity-texture-distance decomposition)

· AI department manager in the flab. Conduct Research and guide students.

· Release the first thermal hyperspectral image restoration methods based on Radiative Transfer Equation, **HAIR**.

· Release the first temperauture-emissivity-texture decomposition dataset and benchmark, **TeX-1500**.

· Release the first sparse-active thermal texture imaging framework that isolates reflection-dominant thermal texture through controlled LWIR illumination and leverages RGB-domain pretrained representations for reconstruction, **T²exture**.

## Education

**Wuhan University** Sep. 2025 - Mar. 2026

*Graduate Study in State Key Laboratory of Surveying and Mapping Remote Sensing Information Engineering (LIESMARS), Major in Communication and Information Systems* *Supervisors: Wei He*

- GPA: 90.79% (among 6 postgraduate courses). recipient of the First-Class Graduate Scholarship.
- Voluntarily withdrew after consultation with my supervisor to pursue a more suitable research direction in physical AI.

**Jilin University** Sep. 2021 - Jun. 2025

*BSc. Major in Communication Engineering*

- GPA: 90.58%. Rank: 12/283 (Outstanding graduate, top 5%), 508/710 in CET6. (German and Russian study for 2 years)
- Outstanding Undergraduate Thesis (top 3%).
- Outstanding Student of the School (*4). First-Class Scholarship, Second-Class Scholarship (*3)

## Publications

**[1] Cheng Dai,** Jiale Lin, Hongyi Xu, Bingxuan Song, Ziyang Xie, Fanglin Bao

TeX-1500: A Paired Real-World LWIR Hyperspectral Dataset and Benchmark for Temperature-Emissivity-Texture Decomposition. [Paper], [Homepage], [huggingface] (arxiv version, published in the future)

**[2] Cheng Dai,** Jiale Lin, Bingxuan Song, Yifei Chen, Jiashuo Chen, Xin Yuan, and Fanglin Bao

HAIR: HADAR-Based Thermal Infrared Hyperspectral Image Restoration. [Paper], [Homepage] (IEEE **TIP** under review**)**

**[3]** Jiashuo Chen, **Cheng Dai (co-first author and listed alphabetically)**, and Fanglin Bao

T2exture: Sparse Active Thermal Texture Imaging. [Paper], [Homepage], [huggingface] (**AAAI** under review)

**Note: Independently propose the method, finish main experiments, draw main figures, and write the top-4 sections.**

**[4]** Hongyi Xu, Jiale Lin, **Cheng Dai**, Du Wang, Chenjun Zhao, Jiashuo Chen, Liqin Cao, Yanfei Zhong, Yiyuan She, Fanglin Bao.

Universal computational thermal imaging overcoming the ghosting effect. [Paper] (**Optica** Under review)

## Internship

**Lightwheel, Embodied Assets Group** Dec. 2024 - Feb. 2025

Group Leader: Jian Yang, and Soroush Nasiriany

・ Built tools for embodied simulation, asset construction, data collection, and teleoperation with Blender, MuJoCo, Robosuite, Issacsim and RoboCasa.[Few Contribute Codes, Merged in Final Robocasa]

・ Achievement: develop spacemouse, blender2yaml, yaml2blender plugins, reducing validations cycles 20× Faster.

**Amoon AI, Large Audio Model Department** Apr. 2025 - Sep. 2025

・ Independently collect and label 20k infant-cry audios (update 6 versions), and train or finetune models to classify the baby cries and detect activities. Involved Models: BEATs, Audio-MAE, Wav2vec2, Qwen-audio, Whisper, Conformer.

・ Achievements: reach 67.2% acc in five-class (diaper, hunger, hug, sleepy, gasping) edge deployment device.

## Awards in Competition

・ First Prize in the Provincial Level of the National Undergraduate Mathematics Competition。

・ Honorable Mention in the American Mathematical Competition。

・ Student Leader in Science and Technology Innovation Plan (Changchun Institute of Optics, CAS), **excellent completion**。

・ Student Leader in Provincial-level college student innovation and entrepreneurship competition, successful completion.

## Skills

・ Train and Finetune pretrained Generative Model (Language Model, Vision Language Model, Audio Model .etc.)

・ Improvements to Generative Model (Multi-branch diffusion, Schrodinger Bridge or inference enhancement .etc.) and Interpretability Analysis (i.e. Fisher Information theory, Convex Theory, Derivation of differential equations).

・ Quantum Calculation and Quantum Machine Learning.

・ Model-based Physical Inversion, Numerical Computation, Hyperspectral Image Processing.

・ Time Series Model (i.e. PatchTST, DLinear, TimesFM) and signal processing (i.e. EEG, ECG, Earth vein waves)

**Software:** [**Cuda-Jax**, Latex, Codex, Git, Ubuntu, Python, C++, FFmpeg, Html5, CSS3, Javascript, MATLAB].